6d4e913a-ce91-4b0e-8fb1-d8e541f90aba

<current>

<request>

You are working on an issue in the 'EliasOenal/multimon-ng' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/multimon-ng/multimon-ng (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-01-17T05:14:09.955Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Fix DTMF/ZVEI decoder tests under Wine

DTMF and ZVEI end-to-end tests are currently disabled under Wine in test/run_tests.sh (line 120). They fail when running Windows binaries under Wine, while FLEX/POCSAG tests pass. Native Linux builds work fine for all tests and can be used as a reference.

Symptoms:

DTMF: No decoded output
ZVEI: Partial decodes (ZVEI1: 1, ZVEI1: 4 instead of ZVEI1: 12345)
Likely cause: Tone detection uses floating-point Goertzel filter with cosine table lookups. FLEX/POCSAG use simple rectangular wave
(±amplitude), which works fine under Wine.

Relevant files:

demod_dtmf.c, demod_zvei1.c - decoders
gen_dtmf.c, gen_zvei.c - generators
test/run_tests.sh - currently skips these tests under Wine (line

```
120)
Goal: Re-enable DTMF/ZVEI tests under Wine by fixing the
underlying issue.

Acceptance: DTMF/ZVEI tests pass under WINE_CMD=wine ./test/
run_tests.sh
</problem_statement>
```

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.